Lab 02 – Worksheet

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Note: Assumptions and logics should be explained separately in tasks after the task results.

Task 1.

Provide appropriately commented codes (Mention question part before each part)

Listing 1:

Instruction: bne x22, x23, Else
Machine hex code: 0x017B1663
32-bit binary: 00000001011110110001011001100011

the immediate field extracted from the binary to get PC offset of +12 bytes. This offset matches the value shown by Venus in place of the label Else.

Instruction: beq x0, x0, Exit
Machine hex code: 0x00000463
32-bit binary: 00000000000000000000010001100011

After decoding the SB-format immediate field, the calculated offset is +8 bytes, which matches the offset displayed by Venus for the label Exit.

Listing 2:

Instruction: bne x9, x24, Exit
Machine hex code: 0x01849663
32-bit binary: 00000001100001001001011001100011

The immediate field extracted from the SB-format instruction gives an offset of +12 bytes. This matches the offset shown by Venus after replacing the label Exit.

Instruction: beq x0, x0, Loop
Machine hex code: 0xFE0006E3
32-bit binary: 11111110000000000000011011100011

This is a backward branch. The extracted immediate field, when sign-extended, gives a PC-relative offset of −20 bytes, which matches the offset value shown by Venus for the label Loop.

Instruction	Hex	32-bit Binary	SB type imm (imm[12 | 10:5 | 4:1 | 11])	PC Offset (bytes)
bne x22, x23, Else	0x017B1663	00000001011110110001011001100011	imm[12]=0, imm[10:5]=000000, imm[4:1]=0110, imm[11]=0	+12
beq x0, x0, Exit	0x00000463	00000000000000000000010001100011	imm[12]=0, imm[10:5]=000000, imm[4:1]=0100, imm[11]=0	+8
bne x9, x24, Exit	0x01849663	00000001100001001001011001100011	imm[12]=0, imm[10:5]=000000, imm[4:1]=0110, imm[11]=0	+12
beq x0, x0, Loop	0xFE0006E3	11111110000000000000011011100011	imm[12]=1, imm[10:5]=111111, imm[4:1]=0110, imm[11]=1	-20

Add screenshot of your results (register)

Listing 1:

Machine Code	Basic Code	Original Code
0x017b1663	bne x22 x23 12	bne x22, x23, Else
0x015a09b3	add x19 x20 x21	add x19, x20, x21
0x00000463	beq x0 x0 8	beq x0, x0, Exit
0x415a09b3	sub x19 x20 x21	Else: sub x19,x20, x21

Binary result

$$(017B1663)_{16} = (00000001011110110001011001100011)_2$$

Binary result

$$(00000463)_{16} = (00000000000000000000010001100011)_2$$

Listing 2:

Machine Code	Basic Code	Original Code
0x003b1513	slli x10 x22 3	Loop: slli x10, x22, 3
0x01950533	add x10 x10 x25	add x10, x10, x25
0x00052483	lw x9 0(x10)	lw x9, 0(x10)
0x01849663	bne x9 x24 12	bne x9, x24,Exit
0x001b0b13	addi x22 x22 1	addi x22, x22, 1
0xfe0006e3	beq x0 x0 -20	beq x0, x0, Loop

Binary result

$(01849663)_{16} = (00000001100001001001011001100011)_2$

Binary result

$(FE0006E3)_{16} = (11111110000000000000011011100011)_2$

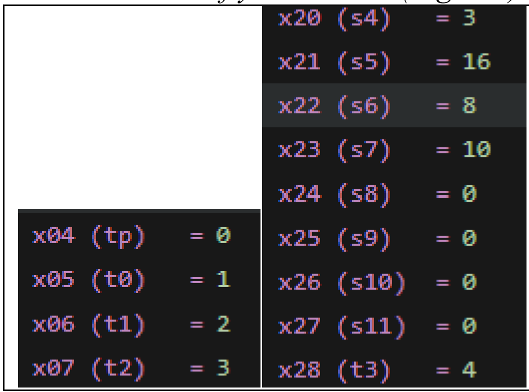
Task 2.

Provide appropriately commented codes (Mention question part before each part)

```
# To Run: setting x = 1 will give 10 + 8 = 18 in x21
addi x20, x0, 3

# Main code
addi x22, x0, 8 # b = 8
addi x23, x0, 10 # c = 10

addi x5, x0, 1 # temp1 = 1
addi x6, x0, 2 # temp2 = 2
addi x7, x0, 3 # temp3 = 3
addi x28, x0, 4 # temp4 = 4

beq x20, x5, case1
beq x20, x6, case2
beq x20, x7, case3
beq x20, x28, case4
beq x0, x0, default

# All Cases
case1:
    add x21, x22, x23
    beq x0, x0, end

case2:
    sub x21, x22, x23
    beq x0, x0, end

case3:
    mul x21, x22, x6
    beq x0, x0, end

case4:
    div x21, x22, x6
```

```
    beq x0, x0, end

default:
    addi x21, x0, 0

end:
```

Add screenshot of your results (register)



Task 3 Provide appropriately commented codes (Mention question part before each part)

```
addi x22, x0, 0 # i = 0
addi x23, x0, 0 # sum = 0
addi x5, x0, 10 #temp1 = 10 for checking loop end
addi x6, x0, 0 # newi = 0 computes newi = 4(i)
sw x21, 0x200(x0)

Loop1:
    beq x22, x5, Initialize
    sw x22, 0x200(x6)
    addi x6, x6, 4
    addi x22, x22, 1
    beq x0, x0, Loop1

Initialize:
    addi x22, x0, 0
    addi x6,  x0, 0

Loop2:
    beq x22, x5, end
    lw x28, 0x200(x6)
    add x23, x23, x28
    addi x22, x22, 1
    addi x6, x6, 4
    beq x0, x0, Loop2

end:
    j end
```

Add screenshots of your results (register & memory both)

```
x05 (t0)   = 10
x06 (t1)   = 40
x22 (s6)   = 10
x23 (s7)   = 45
x28 (t3)   = 9
```

0x00000224	09
0x00000220	08
0x0000021C	07
0x00000218	06
0x00000214	05
0x00000210	04
0x0000020C	03
0x00000208	02
0x00000204	01
0x00000200	00

Task 4 (Challenge)

Provide appropriately commented codes (Mention question part before each part)

```
main:
    li x5,4 #a
    li x6,4 #b
    li x7,0 #initial i=0
    li x10,0x100
outerloop:
    bge x7,x5,end
    li x29,0 #j countrer
    beq x0,x0, innerloop
innerloop:
    bge x29,x6,incrementer
    li x11,4
    mul x11,x11,x29
    add x11,x11,x10
    add x12,x7,x29
    sw x12,0(x11)

    addi x29,x29,1
    beq x0,x0,innerloop
incrementer:
    addi x7,x7,1
    beq x0,x0,outerloop
end:
      j end
```

Add screenshots of your results (register & memory both)

sp (x2)	0x7ffffff0
gp (x3)	0x10000000
tp (x4)	0x00000000
t0 (x5)	0x00000004
t1 (x6)	0x00000004
t2 (x7)	0x00000004
s0 (x8)	0x00000000
s1 (x9)	0x00000000
a0 (x10)	0x00000100
a1 (x11)	0x0000010c
a2 (x12)	0x00000006
t4 (x29)	0x00000004

0x0000010c	06
0x00000108	05
0x00000104	04
0x00000100	03

Assessment Rubric
Lab 02: Implementing Decision Instructions in RISC – V Assembly Language

Task No.	LR 2 Code	LR 5 Results
Task 1	/0	/5
Task 2	/10	/10
Task 3	/15	/10
Task 4	/20	/10
Total Points: 100	**/45**	**/35**
CLO Mapped	CLO 2	

Affective Domain	Rubric	Points	CLO Mapped
AR7	Report Submission & Git Upload	/10 & /10	CLO2

CLO	Total Points	Points Obtained
2	100	
Total	100	

For description of different levels of the mapped rubrics, please refer to the Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics provided here.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model doesnot implement the requiredfunctionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulatio nmodel/network model has some errors and doesnot produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulatio nmodel/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full commandon the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorlyconstructed with erroneousresults. Titles, captions, units are not mentioned. Data is presented in anobscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are notaccurate. Data presentation is not tooclear.	All figures, graphs, tables are correctly drawn but contain minorerrors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captionsand proper units are mentioned. Data presentation is systematic.
AR9	**Report Content/Code Comments**	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots andfigures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plotsand figure with proper label, titles and captions